

RoboSTD: Zero-Shot Single-to-Dual Transfer via Sagittal Mirroring for Bimanual Learning

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Abstract—Bimanual robot manipulation plays an important role in real-world robotic applications, but the high cost of collecting bimanual demonstrations limits the scalable development of bimanual policy learning. Meanwhile, existing single-arm manipulation data are difficult to directly transfer to bimanual systems due to the Single-to-Dual gap in action spaces, coordination requirements, and arm-side spatial preference. To address this issue, we propose RoboSTD, a zero-shot Single-to-Dual data construction method for bimanual robot learning. RoboSTD first applies sagittal-plane mirroring to observations, proprioception, and actions to construct executable contralateral pseudo-demonstrations. It then uses an LLM to generate structured spatio-temporal coordination constraints that guide the reconstruction of pseudo-bimanual supervision across workspace regions and task stages. We evaluate RoboSTD on the AgileX ALOHA platform in both RoboTwin simulation and real-world settings. Experiments demonstrate the effectiveness of RoboSTD for bimanual policy learning. Without requiring additional bimanual demonstrations, RoboSTD increases the average number of collected cups to $2.11\times$ that of the single-arm baseline in the cup collection task. *Project page:* <https://huahungy.github.io/RoboSTD/>

I. INTRODUCTION

Bimanual robot manipulation enables complex tasks that are difficult to efficiently perform with single-arm systems through coordinated use of two arms, making it important for real-world robotic applications [1], [2]. However, compared with single-arm learning, bimanual policy learning faces a more severe data bottleneck. Collecting high-quality bimanual demonstrations requires not only recording the trajectories of both arms, but also maintaining spatial relationships, execution timing, and task-role coordination, making data acquisition substantially more complex and expensive. Previous studies have identified demonstration collection cost and teleoperation burden as major bottlenecks for scalable robot learning [3]–[5]. Therefore, reducing the dependence on real bimanual demonstrations is critical for scalable bimanual robot learning.

Meanwhile, the amount of single-arm manipulation data accumulated by the robot learning community represents a valuable resource for robot learning. However, how to reuse such single-arm data for bimanual learning remains under-explored. This gives rise to a *Single-to-Dual gap*, involving differences in action spaces, coordination requirements, and spatial preferences across arm sides. An overview of this challenge and our approach is shown in Fig. 1. To investigate

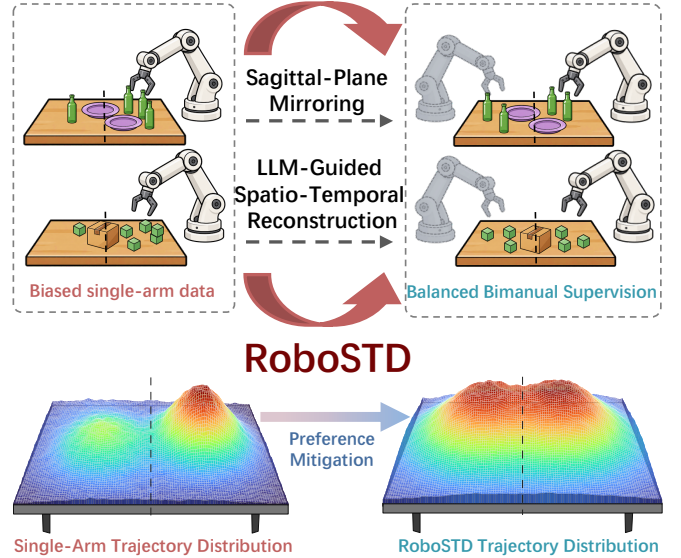


Fig. 1. **Overview and motivation of RoboSTD.** RoboSTD transforms spatially biased single-arm demonstrations into pseudo-bimanual supervision through two stages: sagittal-plane mirroring constructs executable contralateral skills, while LLM-guided spatio-temporal reconstruction coordinates skills through arm assignment, precedence, and conflict constraints, broadening workspace coverage and mitigating arm-side spatial preference.

this phenomenon, we randomly sample 10,000 trajectories from Open X-Embodiment [6] and compute the spatial locations of end-effector events. As shown in Fig. 2(a), only **35.7%** of these events occur in the left workspace, whereas **64.3%** occur in the right workspace, corresponding to a **28.6 percentage-point** spatial imbalance. We further evaluate the pretrained π_0 [7] and the non-pretrained ACT [1] on 10 tasks in RoboTwin 2.0 [8]. As shown in Fig. 2(b), π_0 achieves average success rates of **66.3%** and **75.9%** on left- and right-arm tasks, respectively, corresponding to a **Bias Gap of 9.6 percentage points**. In contrast, ACT achieves **50.0%** and **50.4%**, with a **Bias Gap of only 0.4 percentage points**. The right-side preference observed in both the data distribution and the pretrained policy suggests that upstream spatial imbalance may be reflected in arm-side performance, potentially hindering balanced Single-to-Dual transfer.

To address this gap, we propose **RoboSTD**, a zero-shot Single-to-Dual data construction method for bimanual robot learning. Here, *zero-shot* denotes requiring no additional task-specific bimanual demonstrations, rather than zero-shot policy inference. RoboSTD consists of two stages. First, **sagittal-plane mirroring** transforms observations, prop-

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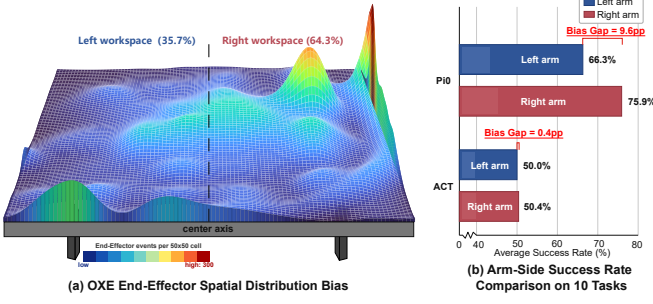


Fig. 2. **Arm-side spatial preference analysis in robot data and policies.** (a) End-effector workspace distribution estimated from Open X-Embodiment trajectories to characterize workspace bias. (b) Left- and right-arm performance comparison of pretrained π_0 and non-pretrained ACT across 10 RoboTwin 2.0 tasks, examining whether data-level spatial imbalance is reflected in downstream policy performance.

oception, and actions [9], [10] to construct executable contralateral pseudo-demonstrations. Second, LLM-guided spatio-temporal reconstruction infers arm assignment, precedence, and conflict constraints from context to compose original and mirrored skills into pseudo-bimanual supervision.

We evaluate RoboSTD using the AgileX ALOHA platform in both the RoboTwin 2.0 simulation environment [8] and real-world settings. In the simulation effectiveness experiments over 10 RoboTwin tasks, policies trained with 100 native bimanual demonstrations per task achieve an average success rate of **67.6%**, while those trained with RoboSTD-generated supervision from only 50 single-arm demonstrations per task achieve **66.5%**. The small **1.1** percentage-point gap shows that RoboSTD can construct effective bimanual supervision from fewer single-arm demonstrations. In the spatial generalization task, RoboSTD extends skills originally limited to a unilateral workspace to the full bimanual workspace, achieving a **2.11 $\times$** increase in the average number of collected cups over the single-arm baseline. In the long-horizon sandwich-making task, RoboSTD further improves the task success rate and average completion, demonstrating stronger temporal coordination. These results demonstrate that sagittal-plane mirroring and LLM-guided spatio-temporal reconstruction can effectively extend single-arm demonstrations into useful bimanual supervision.

The main contributions are summarized as follows:

- We identify and analyze arm-side spatial preferences in robot data and pretrained policies, and reveal their potential impact on balanced bimanual learning.
- We propose RoboSTD, a zero-shot Single-to-Dual data construction method that converts existing single-arm demonstrations into bimanual supervision.
- We demonstrate RoboSTD’s effectiveness for Single-to-Dual transfer, improving workspace coverage and long-horizon coordination in real-world bimanual tasks.

II. RELATED WORK

Unimanual and Bimanual Robot Datasets. Large-scale robot datasets have become an important foundation for learning generalizable manipulation policies. Open X-

Embodiment [6] aggregates large-scale cross-embodiment robot data, while DROID [11] and BridgeData [12] provide diverse real-world manipulation data across tasks and environments. However, most existing large-scale resources are dominated by unimanual interaction. For bimanual manipulation, ALOHA [1] provides a low-cost dual-arm teleoperation platform for collecting coordinated demonstrations, while Mobile ALOHA [2] extends data collection to mobile bimanual manipulation. RoboCOIN [13] further provides real-world bimanual manipulation data for learning and evaluating coordinated policies. In simulation, RoboTwin 2.0 [8] provides scalable bimanual task environments and data-generation pipelines. Despite these efforts, large-scale, high-quality real-world bimanual data remain substantially more difficult and expensive to acquire than unimanual data.

Unimanual and Bimanual Policy Learning. Recent imitation-learning methods have substantially improved robot manipulation through sequence modeling [1], diffusion-based action generation [14], and large-scale generalist or vision-language-action pretraining [7], [15], [16]. Compared with unimanual manipulation, bimanual policy learning must jointly model the actions of both arms together with their spatial and temporal coordination, introducing additional challenges in role assignment and long-horizon execution. ALOHA [1] and Mobile ALOHA [2] demonstrate bimanual policy learning from coordinated demonstrations. These methods highlight the importance of jointly modeling both arms together with their spatial and temporal coordination in complex bimanual tasks.

Robot Data Augmentation and Bimanual Transfer. Another line of work improves data efficiency by augmenting or reusing robot demonstrations. MimicGen [4] and DexMimicGen [5] automatically generate demonstrations from a small number of source demonstrations, while D-CODA [17] and ROPA [18] improve the utilization of existing bimanual demonstrations through offline data augmentation. AnyBimanual [19] transfers pretrained unimanual policies to bimanual manipulation, but still relies on additional bimanual demonstrations for adaptation. Other studies exploit morphological symmetry to improve sample efficiency and bilateral generalization [9], [20]. As summarized in Table I, existing approaches differ in their source supervision, dependence on additional bimanual demonstrations, and ability to support Single-to-Dual transfer and spatio-temporal coordination.

III. PROBLEM FORMULATION

We study zero-shot Single-to-Dual transfer: constructing bimanual supervision from existing single-arm demonstrations without collecting additional task-specific bimanual demonstrations. Here, zero-shot refers to the data requirement rather than policy inference. For arm $k \in \{L, R\}$, let

$$\tau^k = \{(o_t, s_t^k, a_t^k)\}_{t=1}^T, \quad (1)$$

and our goal is to transform single-arm demonstrations into executable coordinated supervision for a bimanual policy.

A single-arm trajectory observes only one side of the interaction, whereas bimanual supervision requires the states

TABLE I
COMPARISON OF DATA REUSE AND SINGLE-TO-DUAL TRANSFER METHODS FOR BIMANUAL ROBOT LEARNING

Method	Source Supervision	No Additional Bimanual	Automated Data Generation	Sagittal Mirroring	Single-to-Dual	Spatio-Temporal Supervision
DexMimicGen	Bimanual demos	×	✓	×	×	○
D-CODA	Bimanual demos	×	✓	×	×	×
ROPA	Bimanual demos	×	✓	×	×	×
AnyBimanual	Pretrained unimanual policy	×	×	×	✓	○
RoboSTD	Single-arm demos	✓	✓	✓	✓	✓

Note: ✓: explicitly supported; ○: partially or implicitly supported; ×: not supported.

and actions of both arms together with spatio-temporal coordination. Given only τ^k , the contralateral state $s_t^{\bar{k}}$, action $a_t^{\bar{k}}$, and coordination annotation c_t are unavailable, where c_t denotes the language-form annotation of the current subtask, arm role, and temporal stage. The action space expands from

$$\mathcal{A}_{\text{single}} = \mathcal{A}^k, \quad \mathcal{A}_{\text{bi}} = \mathcal{A}^L \times \mathcal{A}^R, \quad (2)$$

making direct reuse of single-arm demonstrations generally insufficient for coordinated bimanual policy learning.

Recovering valid bimanual supervision from single-arm data requires more than flipping observations or copying robot commands. Differences in coordinate frames, joint axes, and mounting configurations can break the correspondence between observations, proprioceptive states, and actions across the two arms. Therefore, Single-to-Dual transfer requires a cross-arm mapping that preserves consistency.

Even with valid mirrored skills, directly combining original and mirrored trajectories does not guarantee coordinated bimanual behavior. The two arms may enter overlapping workspace regions or execute incompatible actions at the same stage. Long-horizon tasks require arm assignment and sequencing to avoid spatial conflicts and temporal ambiguity.

Given only $\mathcal{D}_{\text{single}}$, Single-to-Dual aims to construct

$$\tilde{\mathcal{D}}_{\text{bi}} = \mathcal{F}(\mathcal{D}_{\text{single}}), \quad (3)$$

such that the resulting supervision is physically executable and spatio-temporally coordinated for bimanual policy learning. This leads to two core motivations:

Motivation 1: Physically Valid Cross-Arm Transfer.

How can single-arm demonstrations be mapped across arms with geometric and control consistency?

Motivation 2: Coordinated Spatio-Temporal Composition. How can transferred skills be organized without workspace conflicts or temporal ambiguity?

IV. METHOD

RoboSTD constructs pseudo-bimanual supervision from single-arm demonstrations, robot geometry, and task context, as shown in Fig. 3(a). Let $\mathcal{D}_{\text{single}}$ denote the source dataset, $\mathcal{U}$ the robot URDF (Unified Robot Description Format), and $\mathcal{C} = (\ell, \mathcal{O}, \mathcal{W})$ the task context. The two stages, sagittal-plane mirroring and LLM-guided spatio-temporal reconstruction, jointly enable skill transfer and coordination without

collecting additional task-specific bimanual demonstrations (Fig. 3(b–d)). Algorithm 1 summarizes the pipeline.

A. Sagittal-Plane Mirroring

Motivated by morphological symmetry [9] and multimodal reflection [10], RoboSTD constructs contralateral supervision using a platform-specific mapping. Given a trajectory $\tau^k = \{(o_t^k, s_t^k, a_t^k)\}_{t=1}^T$ from arm $k \in \{L, R\}$, o_t^k , s_t^k , and a_t^k denote the visual observation, proprioceptive state, and action at timestep t , respectively, and T is the trajectory length. The pseudo-demonstration for the opposite arm $\bar{k}$ is defined

$$\tilde{\tau}^{\bar{k}} = \mathcal{M}(\tau^k) = \left\{ (\tilde{o}_t^{\bar{k}}, \tilde{s}_t^{\bar{k}}, \tilde{a}_t^{\bar{k}}) \right\}_{t=1}^T, \quad (4)$$

where a tilde denotes a mirrored quantity. The operator $\mathcal{M}$ exchanges arm identity and applies $\tilde{o}_t^{\bar{k}} = \mathcal{M}_o(o_t^k)$, $\tilde{s}_t^{\bar{k}} = \mathcal{M}_s(s_t^k)$, and $\tilde{a}_t^{\bar{k}} = \mathcal{M}_a(a_t^k)$. Here, $\mathcal{M}_o$, $\mathcal{M}_s$, and $\mathcal{M}_a$ are the observation, state, and action mappings for sagittal reflection, preserving their physical correspondence under the symmetry. Applying $\mathcal{M}$ to $\mathcal{D}_{\text{single}}$ yields $\tilde{\mathcal{D}}_{\text{single}}$.

In a robot-centered frame with its origin on the sagittal plane, the reflection matrix $\mathbf{S}$ satisfies $\mathbf{S} = \mathbf{S}^T = \mathbf{S}^{-1}$ and $\det(\mathbf{S}) = -1$. Let $\mathbf{p}$ and $\mathbf{R}$ denote the end-effector position and orientation matrix. With matching world- and end-effector-frame reflection conventions, their mirrored counterparts are $\tilde{\mathbf{p}} = \mathbf{S}\mathbf{p}$ and $\tilde{\mathbf{R}} = \mathbf{S}\mathbf{R}\mathbf{S}$. Writing $\mathbf{R}(\mathbf{v}, \theta)$ for a rotation by angle θ about unit rotation axis $\mathbf{v}$ gives

$$\mathbf{S}\mathbf{R}(\mathbf{v}, \theta)\mathbf{S} = \mathbf{R}(\mathbf{S}\mathbf{v}, -\theta), \quad (5)$$

which determines the corresponding rotation-axis and angle-sign changes required for a physically mirrored orientation.

For kinematically symmetric arms with matched joint ordering and mirror-compatible zero configurations, we use

$$\tilde{\mathbf{q}}^{\bar{k}} = \mathbf{\Lambda}\mathbf{q}^k, \quad \mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n), \quad (6)$$

where $\mathbf{q}^k \in \mathbb{R}^n$ contains the n revolute-joint angles of arm k , and $\lambda_i \in \{+1, -1\}$ determines whether joint i preserves or reverses its angle sign. Let $\mathbf{v}_i^k$ and $\mathbf{v}_i^{\bar{k}}$ denote corresponding unit joint axes at reference configurations in a common frame. The joint signs follow from the reflected-axis relation:

$$\lambda_i = \begin{cases} +1, & \mathbf{v}_i^{\bar{k}} = -\mathbf{S}\mathbf{v}_i^k, \\ -1, & \mathbf{v}_i^{\bar{k}} = \mathbf{S}\mathbf{v}_i^k. \end{cases} \quad (7)$$

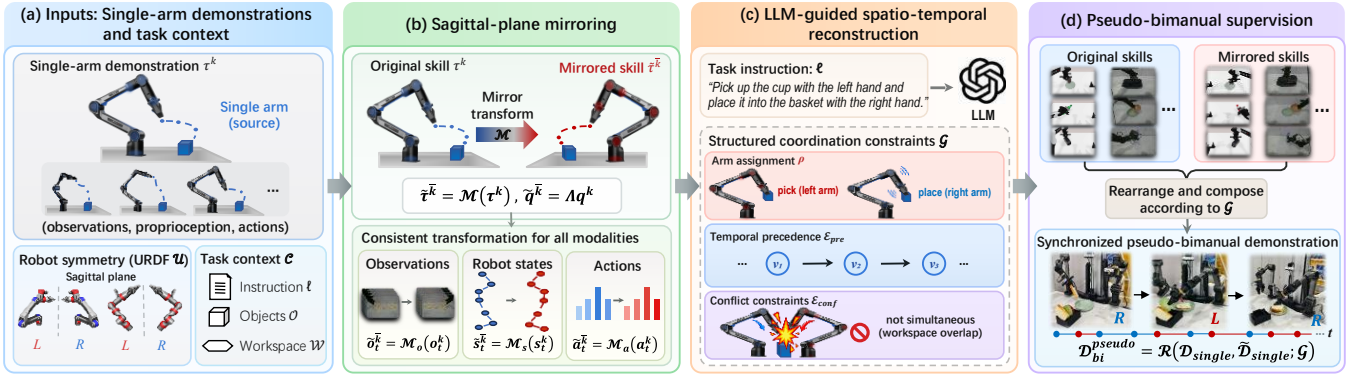


Fig. 3. **Overview of RoboSTD.** (a) Inputs include single-arm demonstrations, robot geometry, and task context. (b) Sagittal-plane mirroring jointly transforms observations, proprioception, and actions to generate contralateral pseudo-demonstrations. (c) An LLM infers arm assignments, precedence, and conflict constraints. (d) These constraints guide the composition of original and mirrored skills into pseudo-bimanual supervision.

The special case $\Lambda = -\mathbf{I}$, with $\mathbf{I}$ the n -dimensional identity matrix, yields $\tilde{\mathbf{q}}^k = -\mathbf{q}^k$. RoboSTD derives Λ from joint-axis definitions and mounting configuration and applies it consistently to the corresponding joint states and actions.

B. LLM-Guided Spatio-Temporal Reconstruction

Sagittal-plane mirroring provides executable skills for both arms, but independent single-arm trajectories do not specify how these skills should be assigned and coordinated in a bimanual task. RoboSTD therefore uses an LLM to infer structured spatio-temporal constraints from task context $\mathcal{C} = (\ell, \mathcal{O}, \mathcal{W})$. Here, ℓ is the language instruction, $\mathcal{O}$ contains object identities and locations, and $\mathcal{W}$ specifies workspace, reachability, and safety constraints for both arms.

We first decompose the task into N manipulation units $\mathcal{V} = \{v_i\}_{i=1}^N$, where each v_i denotes a segment associated with an available original or mirrored skill. Given $\mathcal{V}$ and task context $\mathcal{C} = (\ell, \mathcal{O}, \mathcal{W})$, the LLM generates the following structured coordination constraints, as shown in Fig. 3(c):

$$\mathcal{G} = \text{LLM}(\mathcal{V}, \ell, \mathcal{O}, \mathcal{W}) = (\rho, \mathcal{E}_{pre}, \mathcal{E}_{conf}), \quad (8)$$

where $\rho = \{\rho_i\}_{i=1}^N$ contains the arm assignments, with $\rho_i \in \{L, R\}$ identifying the arm responsible for unit v_i . The directed relation $(v_i, v_j) \in \mathcal{E}_{pre}$ requires v_i to precede v_j , while $\mathcal{E}_{conf}$ contains pairs of units that should not execute simultaneously due to workspace overlap or task conflicts.

In implementation, we use GPT-4.1 [21] as the planner. It returns ρ , $\mathcal{E}_{pre}$, and $\mathcal{E}_{conf}$ via a predefined JSON schema, which is automatically validated before reconstruction.

As shown in Fig. 3(d), the rearrangement operator $\mathcal{R}$ uses these constraints to compose original and mirrored demonstrations into the pseudo-bimanual dataset:

$$\mathcal{D}_{bi}^{pseudo} = \mathcal{R}(\mathcal{D}_{single}, \tilde{\mathcal{D}}_{single}; \mathcal{G}). \quad (9)$$

For each unit, $\mathcal{R}$ selects the original or mirrored skill according to ρ_i , orders the selected skills using $\mathcal{E}_{pre}$, and temporally separates conflicting units in $\mathcal{E}_{conf}$. During a single-arm stage, the inactive arm maintains its current configuration. Both arm sequences are then aligned on a common timeline, yielding pseudo-bimanual samples $(o_t, s_t^L, s_t^R, a_t^L, a_t^R)$.

Each reconstructed stage is further assigned a language-form annotation c_t encoding its subtask, arm role, and temporal stage. We augment the task instruction as $\ell_t^{coord} = \ell \oplus c_t$, where $\oplus$ denotes textual composition. The resulting supervision trains the language-conditioned bimanual policy

$$\pi(a_t^L, a_t^R \mid o_t, s_t, \ell_t^{coord}), \quad (10)$$

where $s_t = (s_t^L, s_t^R)$ is the combined proprioceptive state. RoboSTD therefore modifies the training supervision without changing the downstream policy architecture.

C. Algorithm Summary

Algorithm 1 RoboSTD Data Construction

Require: Single-arm dataset $\mathcal{D}_{single}$; robot URDF $\mathcal{U}$ and mounting configuration; task context $\mathcal{C} = (\ell, \mathcal{O}, \mathcal{W})$
Ensure: Pseudo-bimanual dataset $\mathcal{D}_{bi}^{pseudo}$

- 1: Determine $\mathbf{S}$ from robot geometry and mounting
- 2: Derive Λ from common-frame joint axes using Eq. 7
- 3: Initialize $\tilde{\mathcal{D}}_{single} \leftarrow \emptyset$
- 4: **for** each $\tau^k \in \mathcal{D}_{single}$ **do**
- 5: $\tilde{\tau}^k \leftarrow \mathcal{M}(\tau^k; \mathbf{S}, \Lambda)$
- 6: $\tilde{\mathcal{D}}_{single} \leftarrow \tilde{\mathcal{D}}_{single} \cup \{\tilde{\tau}^k\}$
- 7: **end for**
- 8: Decompose task into units $\mathcal{V} = \{v_i\}_{i=1}^N$
- 9: Generate coordination constraints $\mathcal{G} \leftarrow \text{LLM}(\mathcal{V}, \mathcal{C})$
 $\mathcal{G} = (\rho, \mathcal{E}_{pre}, \mathcal{E}_{conf})$
- 10: Rearrange skills using $\mathcal{G}$ and hold inactive arm poses
- 11: Construct $\mathcal{D}_{bi}^{pseudo} \leftarrow \mathcal{R}(\mathcal{D}_{single}, \tilde{\mathcal{D}}_{single}; \mathcal{G})$
- 12: Add stage annotation c_t to the language condition ℓ
- 13: **return** $\mathcal{D}_{bi}^{pseudo}$

V. EXPERIMENTS

In this section, we comprehensively evaluate RoboSTD in simulation and real-world environments to answer research questions on effectiveness, coordination, and bias mitigation:

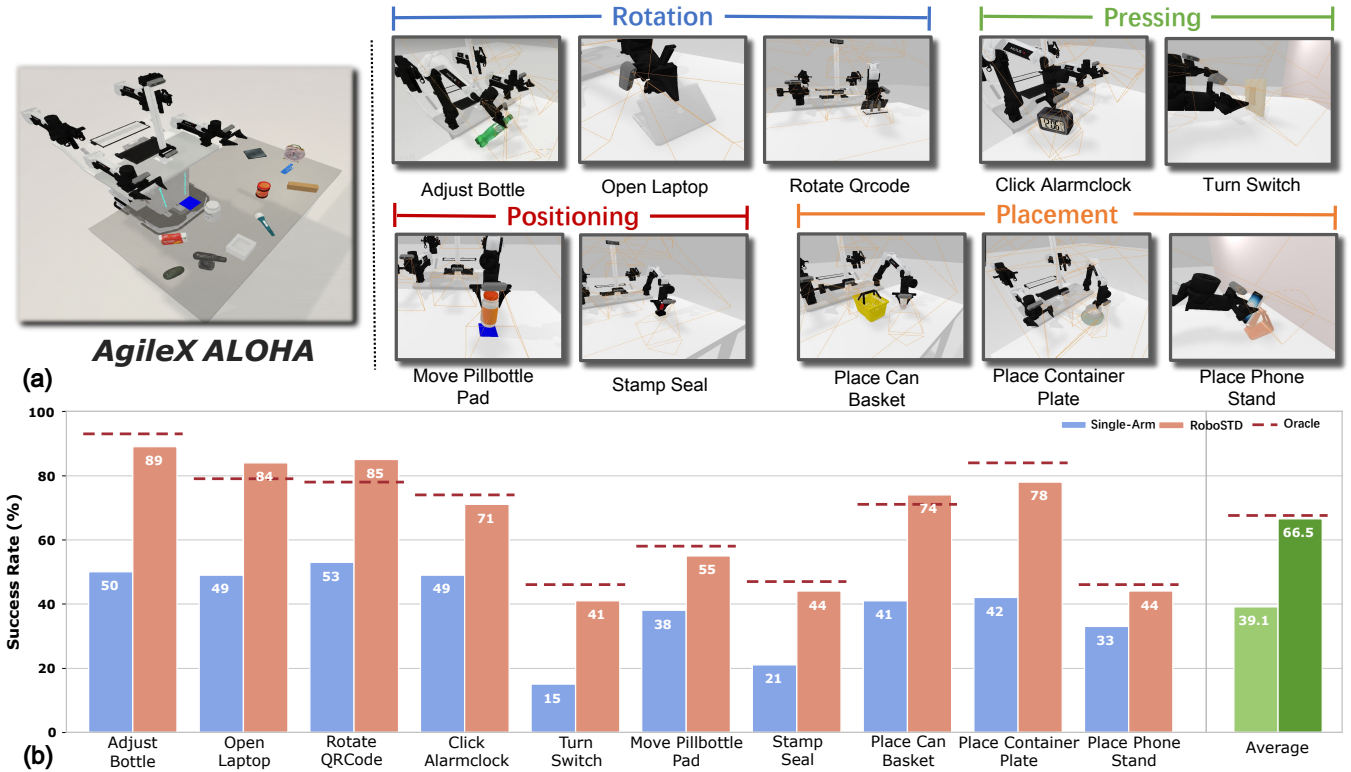


Fig. 4. **RoboSTD evaluation on 10 RoboTwin tasks.** (a) Dual-arm AgileX ALOHA platform and 10 manipulation tasks covering rotation, pressing, positioning, and placement. (b) Task-wise success rates of π_0 under Single-Arm, RoboSTD, and Oracle supervision settings.

- Q1.** Can RoboSTD-generated pseudo-bimanual supervision effectively support bimanual policy learning?
Q2. Can LLM-guided spatio-temporal reconstruction improve bimanual coordination?
Q3. Can RoboSTD mitigate arm-side spatial preference during fine-tuning of pretrained robot policies?

A. Experimental Setup

We evaluate RoboSTD in simulation and real-world environments using the dual-arm AgileX ALOHA platform [1].

Simulation Environment. Simulation experiments are conducted in RoboTwin 2.0 [8] using an AgileX ALOHA robot with two 6-DoF arms. We select 10 manipulation tasks covering rotation, pressing, positioning, and placement. For the effectiveness study, the Single-Arm and RoboSTD settings use the same 50 source single-arm demonstrations per task, while the Oracle setting uses 100 directly collected native bimanual demonstrations per task. For the arm-side spatial preference study, we use a separate set of 1,000 original single-arm demonstrations, with 100 demonstrations for each of the 10 RoboTwin tasks, and evaluate three tasks with the largest right-to-left gaps in the preliminary analysis.

Real-World Environment. We evaluate five real-world bimanual tasks on the same AgileX ALOHA platform. Three low-coordination tasks, *Bowl Placement*, *Towel Storage*, and *Flower Arrangement*, evaluate rigid-object placement, deformable-object manipulation, and precision placement, respectively. Two high-coordination tasks, *Cup Collection*

and *Sandwich Making*, further evaluate spatial coordination and long-horizon temporal coordination. For each task, we collect 50 single-arm demonstrations for RoboSTD and 50 native bimanual demonstrations for the Oracle reference.

Data Configurations. We compare three supervision settings where applicable. *Single-Arm* directly uses the original source single-arm demonstrations; *RoboSTD* constructs pseudo-bimanual supervision from the same source demonstrations; and *Oracle* uses directly collected native bimanual demonstrations as reference supervision. For the high-coordination real-world tasks, we additionally evaluate *RoboSTD w/o LLM*, which retains sagittal-plane mirroring and spatial rearrangement but removes the LLM-generated arm assignment, precedence, and conflict constraints. Original and mirrored skills are composed using the default execution order without explicit spatio-temporal coordination.

Policies. We use π_0 [7] for the RoboTwin experiments and $\pi_{0.5}$ [22] for all real-world experiments in this study. For high-coordination tasks, c_t is incorporated into the language condition, leveraging $\pi_{0.5}$'s semantic subtask conditioning.

Training Details. All policies are fine-tuned with a batch size of 32. We use a 1,000-step warmup followed by a learning-rate decay schedule from 2.5×10^{-5} to 2.5×10^{-6} with 30,000 decay steps. Training is performed on an NVIDIA A800 GPU with 80 GB memory. All configurations use identical optimization settings to ensure fairness.

Evaluation Metrics. Task success rate (SR) is reported for all RoboTwin tasks and applicable real-world tasks. For *Cup Collection*, we additionally report the average number of

collected cups. For *Sandwich Making*, we report task success rate and average completion level. Arm-side performance imbalance is measured by Bias Gap = $SR_{\text{right}} - SR_{\text{left}}$, where a positive value indicates a right-side performance advantage and a smaller gap indicates more balanced arm-side performance. To quantify execution-level spatial preference, we project end-effector trajectories onto the ΔX - ΔZ workspace plane and define Density Gap as follows:

$$\text{Density Gap} = \frac{\bar{\rho}_R - \bar{\rho}_L}{\bar{\rho}_L} \times 100\%, \quad (11)$$

where $\bar{\rho}_R$ and $\bar{\rho}_L$ denote the mean trajectory densities in the right and left workspaces, respectively. A smaller positive value indicates more balanced left-right workspace coverage.

B. Effectiveness of RoboSTD for Bimanual Learning

To answer **Q1**, we evaluate whether pseudo-bimanual supervision constructed by RoboSTD can effectively support bimanual policy learning on 10 manipulation tasks from RoboTwin 2.0 [8]. As shown in Fig. 4(a), the selected tasks cover four representative manipulation categories, including rotation, pressing, positioning, and placement, providing diverse object interactions and workspace configurations for evaluating Single-to-Dual transfer. The Single-Arm and RoboSTD settings use the same 50 source single-arm demonstrations per task, whereas the Oracle reference uses 100 directly collected native bimanual demonstrations per task.

As shown in Fig. 4(b), the Single-Arm setting achieves an average success rate of only **39.1%**, reflecting the difficulty of directly reusing single-arm demonstrations for bimanual execution. In contrast, RoboSTD raises the average success rate to **66.5%**, closely approaching the **67.6%** achieved by the Oracle setting. The difference between RoboSTD and Oracle is only **1.1 percentage points (pp)**, despite RoboSTD using 50 single-arm source demonstrations instead of 100 directly collected bimanual demonstrations. The close performance to Oracle indicates that RoboSTD-generated pseudo-bimanual supervision provides effective training signals comparable to native bimanual demonstrations.

The task-wise results in Fig. 4(b) further show that RoboSTD substantially outperforms the Single-Arm setting across all 10 tasks while remaining close to the Oracle reference overall. On several tasks, including *Open Laptop*, *Place Can Basket*, and *Rotate QRCode*, RoboSTD even achieves higher success rates than the native bimanual reference. These results suggest that sagittal-plane mirroring and reconstruction retain the visual, proprioceptive, and action correspondences required for Single-to-Dual transfer. Overall, RoboSTD provides effective and data-efficient bimanual supervision without requiring additional task-specific bimanual demonstrations for policy training.

C. Bimanual Manipulation Across Coordination Levels

We further evaluate RoboSTD on five real-world bimanual tasks using the dual-arm AgileX ALOHA platform [1] and $\pi_{0.5}$ [22]. As shown in Fig. 5, the tasks are organized according to their inter-arm coordination requirements.

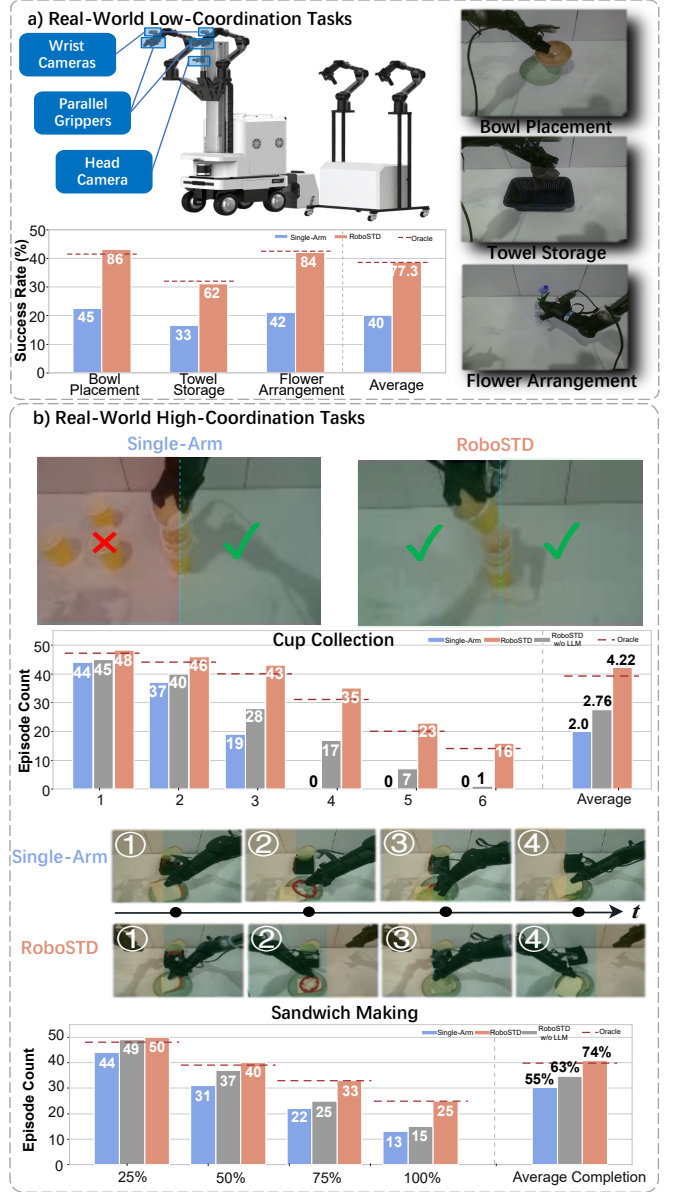


Fig. 5. **Real-world evaluation of RoboSTD across coordination levels.** (a) Low-coordination tasks covering rigid, deformable, and precision manipulation. (b) High-coordination tasks comparing Single-Arm, RoboSTD w/o LLM, RoboSTD, and Oracle to evaluate structured spatio-temporal coordination using task-specific metrics.

The three low-coordination tasks primarily examine whether single-arm skills can be effectively transferred into real-world bimanual supervision, while the two high-coordination tasks further evaluate whether the transferred skills can be spatially and temporally coordinated. For the latter, *RoboSTD w/o LLM* retains the geometric Single-to-Dual construction while removing explicit LLM-generated coordination constraints, isolating the contribution of the spatio-temporal reconstruction and stage-level language supervision introduced in Sec. IV-B for coordinated bimanual execution.

Low-Coordination Bimanual Tasks. As shown in Fig. 5(a), we evaluate *Bowl Placement*, *Towel Storage*, and *Flower Arrangement*, covering rigid-object placement,

deformable-object manipulation, and precision placement, respectively. The Single-Arm setting achieves success rates of **45%**, **33%**, and **42%**, whereas RoboSTD improves them to **86%**, **62%**, and **84%**. Averaged across the three tasks, the success rate increases from **40.0%** to **77.3%**, while remaining close to the Oracle reference. These results further demonstrate that RoboSTD-generated pseudo-bimanual supervision can support real-world bimanual policy learning across tasks with relatively limited inter-arm coordination requirements across diverse object types.

High-Coordination Bimanual Tasks. We next consider *Cup Collection* and *Sandwich Making*, which require explicit spatial and temporal coordination between the two arms. As shown in Fig. 5(b), *RoboSTD w/o LLM* consistently improves over Single-Arm supervision but remains below the full RoboSTD, indicating that sagittal-plane transfer and spatial rearrangement alone are insufficient for strongly coordinated bimanual behaviors. In *Cup Collection*, the average number of collected cups increases from **2.00** with Single-Arm supervision to **2.76** with *RoboSTD w/o LLM*, and further to **4.22** with RoboSTD, slightly exceeding the Oracle result of **3.92**. This highlights the benefit of arm assignment and conflict constraints for organizing transferred skills across the shared workspace. Similarly, in *Sandwich Making*, the average completion level improves from **55%** to **63%** and then to **74%**, compared with **72.5%** for Oracle. The complete-task success rate increases from **26%** to **30%** and finally to **50%**, matching the Oracle success rate of **50%**. These results demonstrate that LLM-guided spatio-temporal reconstruction provides coordination benefits beyond geometric Single-to-Dual transfer, particularly for shared workspaces and long-horizon multi-stage tasks where arm assignment and execution ordering are essential for coordinated bimanual behavior.

Overall, the real-world results reveal two complementary roles of RoboSTD. Sagittal-plane mirroring and spatial rearrangement provide transferable bimanual supervision, while LLM-guided spatio-temporal reconstruction further organizes transferred skills through arm assignment, precedence, and conflict constraints. On the high-coordination tasks, RoboSTD slightly exceeds Oracle in average cup collection (**4.22** vs. **3.92**) and sandwich completion (**74%** vs. **72.5%**), while matching its task success rate (**50%**). These results validate RoboSTD for real-world Single-to-Dual transfer (Q1) and the role of structured spatio-temporal reconstruction in bimanual coordination (Q2).

D. Mitigating Arm-Side Spatial Preference

To answer Q3, we evaluate whether RoboSTD can alleviate right-side spatial preference in pretrained robot policies. We select *Click Alarmclock*, *Move Pillbottle Pad*, and *Rotate QRCode*, the three RoboTwin tasks with the largest right-to-left performance gaps in our preliminary 10-task evaluation. For the Single-Arm baseline, we fine-tune pretrained π_0 on **1,000** original single-arm demonstrations, with 100 demonstrations per task. For RoboSTD, the same 1,000 source demonstrations are converted into pseudo-bimanual supervision to train a RoboSTD-based model, which is then

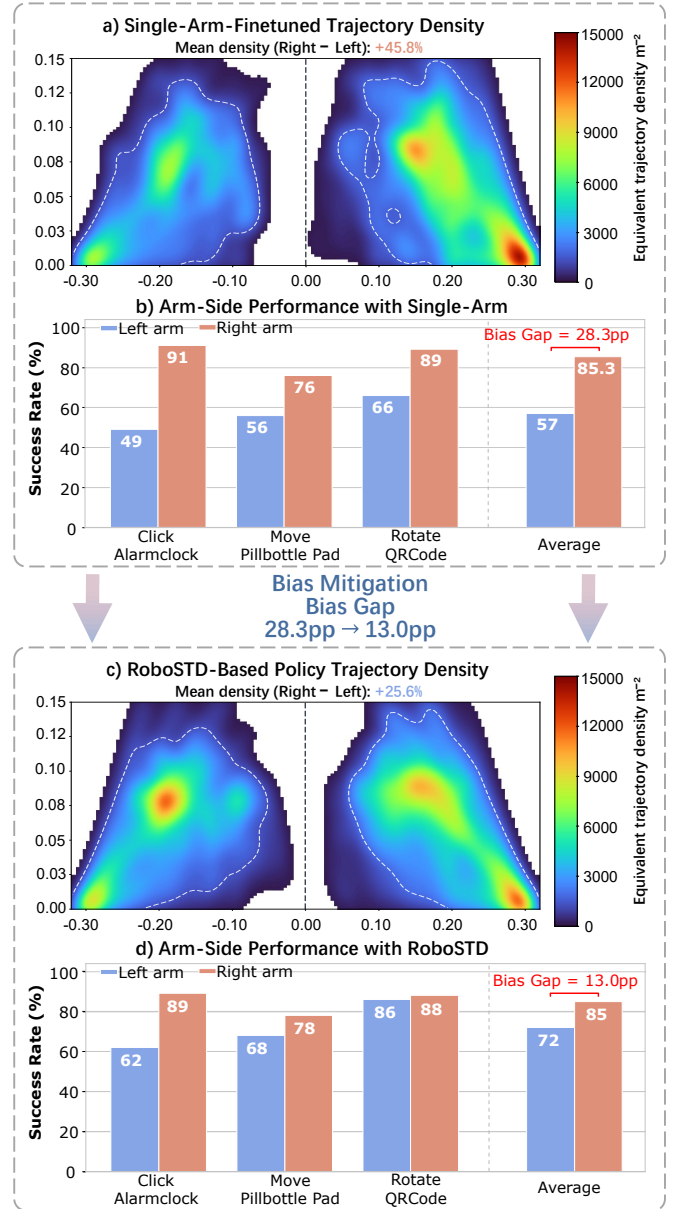


Fig. 6. **RoboTwin evaluation of arm-side spatial preference and its mitigation with RoboSTD.** (a-b) End-effector trajectory density and arm-side success rates of the Single-Arm-finetuned baseline. (c-d) Corresponding results of the RoboSTD-based policy on the three selected RoboTwin tasks.

adapted to the three tasks using their task-specific single-arm demonstrations. As shown in Fig. 6, we compare the two training settings from trajectory-distribution and arm-side performance perspectives to examine whether RoboSTD improves workspace balance and arm-side task performance.

As shown in Fig. 6(a,c), the Single-Arm-finetuned policy exhibits a clear right-side preference in its executed trajectories, with a Density Gap of **45.8%** on the ΔX - ΔZ workspace plane. In contrast, the RoboSTD-based policy reduces the Density Gap to **25.6%**, resulting in more balanced left-right workspace coverage. The redistributed trajectory density further shows that execution is no longer concentrated mainly in the originally preferred right workspace, but

extends more evenly toward the underrepresented left side. This suggests that RoboSTD broadens the spatial utilization of transferred skills across both arm workspaces, yielding more symmetric workspace coverage and reducing the concentration of execution on the originally preferred side.

A consistent trend is observed in task performance (Fig. 6(b,d)). Under Single-Arm supervision, the average right- and left-arm success rates are **85.3%** and **57.0%**, respectively, corresponding to a Bias Gap of **28.3 pp**. With the RoboSTD-based policy, the average left-arm success rate increases to **72.0%**, while right-arm performance remains nearly unchanged at **85.0%**, reducing the Bias Gap to **13.0 pp**. This reduction mainly results from improved left-arm capability rather than degraded right-arm performance. Together with the trajectory-density results, these findings show more balanced arm-side behavior at execution and task-performance levels, supporting RoboSTD in mitigating arm-side spatial preference in pretrained robot policies.

VI. CONCLUSION

In this paper, we presented RoboSTD, a zero-shot Single-to-Dual data construction framework for data-efficient bimanual robot learning. RoboSTD constructs pseudo-bimanual supervision through sagittal-plane mirroring and LLM-guided spatio-temporal reconstruction. The former transfers single-arm skills to the contralateral workspace, while the latter organizes original and mirrored skills through arm assignment, precedence, and conflict constraints. Experiments in simulation and real-world environments demonstrate that RoboSTD can mitigate arm-side spatial preference in robot policies, provide effective pseudo-bimanual supervision, expand single-arm skills to broader bimanual workspaces, and improve long-horizon coordination. These results show that RoboSTD provides an effective way to reuse abundant single-arm demonstrations and alleviate the data bottleneck in bimanual robot learning. More broadly, our results suggest that geometric symmetry can serve as a practical bridge between unimanual and bimanual learning when combined with structured spatial and temporal supervision. This provides a promising direction for scaling bimanual robot learning without relying on large amounts of newly collected bimanual demonstrations for training.

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APPENDIX I

DERIVATION OF MIRROR-CONSISTENT JOINT MAPPING

This appendix provides the detailed derivation of the joint-space mirror mapping used in RoboSTD. The derivation explains the origin of the platform-specific sign matrix Λ used in the main text.

Let the configuration space of an n -DoF robot arm be

$$\mathcal{Q} \subset \mathbb{R}^n. \quad (12)$$

The joint state of the source arm, e.g., the right arm, is denoted as

$$\mathbf{q}^R = [q_1^R, q_2^R, \dots, q_n^R]^T, \quad (13)$$

and the corresponding target-arm state, e.g., the left arm, is denoted as

$$\mathbf{q}^L = [q_1^L, q_2^L, \dots, q_n^L]^T. \quad (14)$$

We assume that corresponding joints of the two arms share consistent actuation definitions, and that when both arms are at their zero configurations, the overall kinematic structure is symmetric with respect to the robot sagittal plane. Let the spatial reflection with respect to the sagittal plane be represented by an orthogonal matrix

$$\mathbf{S} \in \mathbb{R}^{3 \times 3}, \quad (15)$$

where

$$\mathbf{S} = \mathbf{S}^T = \mathbf{S}^{-1}, \quad \det(\mathbf{S}) = -1. \quad (16)$$

For a point $\mathbf{p}$ in task space, its mirrored position is

$$\tilde{\mathbf{p}} = \mathbf{S}\mathbf{p}. \quad (17)$$

For an end-effector pose $\mathbf{T} = (\mathbf{R}, \mathbf{p})$, where $\mathbf{R}$ denotes the orientation and $\mathbf{p}$ denotes the position, the mirrored pose is given by

$$\tilde{\mathbf{T}} = (\mathbf{S}\mathbf{R}\mathbf{S}, \mathbf{S}\mathbf{p}). \quad (18)$$

Thus, both position and orientation are transformed under the same spatial reflection rule.

The key issue in joint-space mapping is the reflection property of rotational axes. Let $\mathbf{R}(\mathbf{v}, \theta) \in SO(3)$ denote a rotation matrix corresponding to a rotation of angle θ around a unit axis $\mathbf{v}$. Under spatial reflection, we have

$$\mathbf{S}\mathbf{R}(\mathbf{v}, \theta)\mathbf{S} = \mathbf{R}(\mathbf{S}\mathbf{v}, -\theta). \quad (19)$$

This property indicates that reflection changes not only the rotation axis but also the sign of the rotation angle. Therefore, whether a mirrored joint angle should preserve or flip its sign depends on the geometric relationship between the original joint axis and its reflected axis.

A. Physically Mirrored Arms

We first consider the case where the left and right arms are physical mirror copies of each other. In this setting, the i -th joint axis of the left arm, $\mathbf{v}_i^L$, is the spatial reflection of the corresponding right-arm joint axis $\mathbf{v}_i^R$:

$$\mathbf{v}_i^L = \mathbf{S}\mathbf{v}_i^R. \quad (20)$$

To generate mirror-symmetric motion, the left-arm rotation must match the reflected right-arm rotation:

$$\mathbf{R}(\mathbf{v}_i^L, q_i^L) = \mathbf{R}(\mathbf{S}\mathbf{v}_i^R, -q_i^R). \quad (21)$$

Since $\mathbf{v}_i^L = \mathbf{S}\mathbf{v}_i^R$, the two rotation axes are identical. Therefore, the joint angle should satisfy

$$q_i^L = -q_i^R. \quad (22)$$

Thus, for physically mirrored arms, the joint-space mapping can be written as

$$\mathbf{q}^L = -\mathbf{q}^R. \quad (23)$$

B. Pitch-, Yaw-, and Roll-like Joints in Isomorphic Arms

In practical bimanual robot systems, a more common configuration is to use two identical robot arms mounted symmetrically. In this case, the two arms are not physical mirror copies; instead, they are isomorphic arms with the same mechanical structure. Under symmetric mounting, the corresponding joint axes of the two arms can be approximately treated as identical in the robot coordinate frame:

$$\mathbf{v}_i^L = \mathbf{v}_i^R. \quad (24)$$

However, the theoretically desired mirrored rotation for a symmetric task remains

$$\mathbf{R}(\mathbf{S}\mathbf{v}_i^R, -q_i^R). \quad (25)$$

Therefore, the target-arm joint angle q_i^L must satisfy

$$\mathbf{R}(\mathbf{v}_i^R, q_i^L) = \mathbf{R}(\mathbf{S}\mathbf{v}_i^R, -q_i^R). \quad (26)$$

The sign of q_i^L is determined by how the joint axis $\mathbf{v}_i^R$ changes under sagittal-plane reflection. For clarity, assume that the sagittal plane is the x - z plane of the robot coordinate frame and that the lateral direction is the y -axis. The reflection then flips the y -axis while preserving directions within the x - z plane. This leads to two typical cases.

1) *Pitch-like joints*: A pitch-like joint usually rotates around a lateral axis, which is approximately perpendicular to the sagittal plane. Therefore, its axis changes direction under sagittal-plane reflection:

$$\mathbf{S}\mathbf{v}_i^R = -\mathbf{v}_i^R. \quad (27)$$

Substituting this into the mirrored rotation condition gives

$$\mathbf{R}(\mathbf{v}_i^R, q_i^L) = \mathbf{R}(-\mathbf{v}_i^R, -q_i^R). \quad (28)$$

Using the axis-angle equivalence

$$\mathbf{R}(-\mathbf{v}, -\theta) = \mathbf{R}(\mathbf{v}, \theta), \quad (29)$$

we obtain

$$\mathbf{R}(\mathbf{v}_i^R, q_i^L) = \mathbf{R}(\mathbf{v}_i^R, q_i^R). \quad (30)$$

Hence,

$$q_i^L = q_i^R. \quad (31)$$

Therefore, for pitch-like joints in isomorphic bimanual systems, the mirrored joint angle typically preserves its sign.

2) *Yaw- and roll-like joints:* Yaw-like and roll-like joints usually rotate around axes that lie in, or are parallel to, the sagittal plane. Such axes remain unchanged under sagittal-plane reflection:

$$\mathbf{S}\mathbf{v}_i^R = \mathbf{v}_i^R. \quad (32)$$

Substituting this into the mirrored rotation condition gives

$$\mathbf{R}(\mathbf{v}_i^R, q_i^L) = \mathbf{R}(\mathbf{v}_i^R, -q_i^R). \quad (33)$$

Thus,

$$q_i^L = -q_i^R. \quad (34)$$

Therefore, for yaw-like and roll-like joints in isomorphic bimanual systems, the mirrored joint angle typically flips its sign.

C. Unified Representation

For an isomorphic bimanual system composed of two identical robot arms, the joint-space mirror mapping can be written in a unified form:

$$\mathbf{q}^L = \mathbf{\Lambda}\mathbf{q}^R, \quad (35)$$

where

$$\mathbf{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n), \quad \lambda_i \in \{1, -1\}. \quad (36)$$

The diagonal element λ_i is determined by the relationship between the corresponding joint axis and the sagittal plane:

$$\lambda_i = \begin{cases} +1, & \mathbf{S}\mathbf{v}_i^R = -\mathbf{v}_i^R, \\ -1, & \mathbf{S}\mathbf{v}_i^R = \mathbf{v}_i^R. \end{cases} \quad (37)$$

Equivalently, in common isomorphic bimanual systems, this can be interpreted as

$$\lambda_i = \begin{cases} +1, & \text{pitch-like joint,} \\ -1, & \text{yaw-like or roll-like joint.} \end{cases} \quad (38)$$

It should be noted that the names pitch, yaw, and roll depend on the robot coordinate frame and joint-axis definitions. In practice, RoboSTD does not rely directly on joint names. Instead, $\mathbf{\Lambda}$ is determined from the joint-axis directions in the robot URDF, the robot mounting configuration, or a small number of calibration motions. Once determined, $\mathbf{\Lambda}$ is fixed for the corresponding robot platform and used to transform proprioceptive states and joint-space actions during RoboSTD data construction.